# Quiz Prep: HW 11.2, 11.3, 11.4, 12.1

Class: PHYS 218

The quiz has two problems: one from **11.2 or 11.3**, and one from **11.4 or 12.1**.

I read the Expert TA problem sets for 11.3, 11.4, and 12.1, plus the 11.2 velocity-selector problem. I only saw problem 1 of 11.2, so the 11.2 section is the least certain.

## Equation sheet

### Constants

| Quantity | Value |
|---|---|
| $\mu_0$ (permeability of free space) | $4\pi \times 10^{-7}$ T · m/A |
| $\mu_0/2\pi$ | $2 \times 10^{-7}$ T · m/A |
| $\mu_0/4\pi$ | $1 \times 10^{-7}$ T · m/A |
| $e$ (elementary charge) | $1.602 \times 10^{-19}$ C |
| $m_p$, $m_e$ (proton, electron mass) | $1.673 \times 10^{-27}$ kg, $9.109 \times 10^{-31}$ kg |

### Group 1: HW 11.2 and 11.3 (force on charges and wires)

| Idea | Equation | What the variables mean |
|---|---|---|
| Magnetic force on a moving charge | $\vec{F} = q\vec{v} \times \vec{B}, \;\; F = qvB\sin\theta$ | $q$ = charge of the particle (C, with sign); $\vec{v}$ = its velocity (m/s); $\vec{B}$ = magnetic field (T); $\theta$ = angle between $\vec{v}$ and $\vec{B}$ |
| Lorentz force | $\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$ | $\vec{E}$ = electric field (V/m) from the plates or source, the other symbols as above |
| Velocity selector (net force zero) | $qE = qvB \Rightarrow v = E/B$ | $E$ = electric field strength between the plates; $B$ = magnetic field strength; $v$ = the one speed that passes undeflected |
| Force on a current-carrying wire | $\vec{F} = I\vec{L} \times \vec{B}, \;\; F = ILB\sin\theta$ | $I$ = current in the wire (A); $\vec{L}$ = length of wire inside the field, pointing along the current (m); $\vec{B}$ = field the wire sits in; $\theta$ = angle between $\vec{I}$ and $\vec{B}$ |
| Circular motion in $B$ | $r = \dfrac{mv}{qB}, \;\; T = \dfrac{2\pi m}{qB}$ | $r$ = radius of the circular path; $T$ = time for one orbit; $m$ = particle mass; $v$ = speed (perpendicular to $B$) |
| Work-energy for a rod, no friction | $W = Fd = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2Fd/m}$ | $F$ = magnetic force on the rod; $d$ = distance it slides; $m$ = rod mass; $v$ = final speed, from rest |
| Force per unit field strength on a straight segment | $F/B = IL$ | $I$ = loop current; $L$ = length of the segment that is inside the field |

- A magnetic force on a free charge does no work, since $\vec{F} \perp \vec{v}$.
- No force when $\vec{v} \parallel \vec{B}$ (or $\vec{I} \parallel \vec{B}$).
- Unit vectors: $\hat{x} \times \hat{y} = \hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$, $\hat{z} \times \hat{x} = \hat{y}$. Reverse the order and the sign flips. Dots are out of the page, crosses are into it.
- Right-hand rule: fingers along $\vec{v}$ (or $\vec{I}$), curl toward $\vec{B}$, thumb gives $\vec{F}$ for a positive charge. For a negative charge, flip it.

**Group 2: HW 11.4 and 12.1 (loops, torque, fields from wires)**

| Idea | Equation | What the variables mean |
|---|---|---|
| Force per length between parallel wires | $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$ | $F/l$ = force per unit length (N/m); $I_1, I_2$ = the currents in the two wires; $d$ = distance between them |
| Field of a long straight wire | $B = \frac{\mu_0 I}{2\pi r}$ | $I$ = current in the wire; $r$ = perpendicular distance from the wire to the point where you want $B$ |
| Dipole moment of a coil | $\mu = NIA$ | $N$ = number of turns; $I$ = current in the coil; $A$ = area of one turn ($\pi R^2$ for a circle of radius $R$) |
| Torque on a coil | $\vec{\tau} = \vec{\mu} \times \vec{B}$, $\tau = \mu B \sin\theta$ | $\vec{\mu}$ = dipole moment (points along the coil's axis); $\vec{B}$ = external field; $\theta$ = angle between $\vec{\mu}$ and $\vec{B}$ |
| Potential energy of a coil | $U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos\theta$ | same symbols; $U = 0$ when $\mu \perp B$ |
| Rectangular loop torque | $\tau = IAB \sin\theta$ | $A = L_1 L_2$, the loop's area; $I$ = loop current; $B$ = field |
| Hall voltage | $V = \frac{IBw}{neA}$, round wire: $V = \frac{4IB}{ne\pi d}$ | $I$ = current in the wire; $B$ = field across it; $n$ = free-electron density ($\mathrm{m}^{-3}$); $e$ = electron charge; $w$ = width the voltage is measured across; $A$ = wire cross-section; $d$ = wire diameter |
| Drift speed | $v_d = \frac{I}{neA}$ | $v_d$ = average speed of the charge carriers along the wire |
| Work to move parallel wires from $d$ to $2d$, per length | $w = \frac{\mu_0 I_1 I_2}{2\pi} \ln 2$ | $w$ = work per unit length (J/m); $I_1, I_2$ = currents; $d$ = starting separation |
| Biot-Savart | $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I\, d\vec{l} \times \hat{r}}{r^2}$ | $d\vec{l}$ = a tiny piece of wire along the current; $\hat{r}$ = unit vector from that piece to the point P; $r$ = distance to P |

- Right-hand rule for the field around a wire: thumb along $I$, fingers curl the way $\vec{B}$ circles it.
- Right-hand rule for $\vec{\mu}$: fingers follow the current, thumb gives $\vec{\mu}$.
- The torque turns $\vec{\mu}$ toward $\vec{B}$. Stable position is $\vec{\mu} \parallel \vec{B}$ ($U = -\mu B$, $\tau = 0$).

## What to know, by homework

### HW 11.2 (force on a moving charge, velocity selector)

- **Setup:** a positive charge $+q$ moves right with speed $v$ through a uniform field $\vec{B}$ pointing out of the page. A velocity selector adds an electric field $\vec{E}$ so the particle goes straight.
- Use $\vec{F} = q\vec{v} \times \vec{B}$ for the direction: $v$ right, $B$ out of the page, $+q$ gives a magnetic force **down**. The electric force must cancel it, so it points **up**, and for a positive charge $\vec{E}$ points **up**.
- $E = vB$. The charge and mass cancel, so any particle at $v = E/B$ goes straight.
- Negative charge: both force directions flip, so $\vec{E}$ would point down.

### HW 11.3 (force on wires)

1. **Find $\vec{B}$ from $\vec{I}$ and $\vec{F}$.** $I$ = current in the wire, $F$ = the magnetic force shown. Test each direction for $\vec{B}$ until $\vec{I} \times \vec{B}$ points along $\vec{F}$.
2. **Find $\vec{F}$ from $\vec{I}$ and $\vec{B}$.** $\vec{I} \times \vec{B}$. Antiparallel or parallel gives zero.
3. **Rod on rails.** $m = 0.95$ kg is the rod's mass, $L = 0.55$ m the distance between the rails (the length of rod carrying current), $I = 2.3$ A the current in the rod, $B = 0.15$ T the field pointing up, $d = 0.55$ m the distance the rod slides. $F = ILB = 0.190$ N, $W = Fd = 0.104$ J, $v = \sqrt{2Fd/m} = 0.469$ m/s.

4. **Loop partly in a field.** $I = 4.5$ A is the loop current, $L = 0.200$ m the width of the section inside the field. Forces on the two vertical sides cancel, so only the segment inside matters. $F/B = IL = 0.900$ N/T.

**HW 11.4 (loops, torque, Hall)**

1. **Wire next to a rectangular loop.** The long wire carries one current and the loop carries another (15.0 A and 30.0 A, check which is which in the figure). $L = 0.300$ m is the length of each horizontal side, and $d$ is the distance from the long wire to that side: 7.50 cm for the top side, 17.5 cm for the bottom. The two short sides cancel. The near side wins because it is closer. Net force $2.06 \times 10^{-4}$ N, away from the wire.
2. **Three wires on a triangle.** $d = 0.100$ m is the distance between each pair of wires. Wire A carries 5 A out of the page, B carries 10 A into it, and C carries 20 A into it. Find each pair force $\mu_0 I_i I_j / 2\pi d$, add as vectors, and check the sum over all three wires is zero.
3. **Coil in a field.** $N = 390$ turns, $I = 3.5$ A is the coil current, $R = 0.55$ m is the coil radius, $B = 1.1$ T points right. $\mu = NI\pi R^2 = 1.30 \times 10^3$ A $\cdot$ m$^2$ (points up). $\tau = \mu B = 1.43 \times 10^3$ N $\cdot$ m, into the page, and $U = 0$ since $\mu \perp B$. If $B$ points along $\mu$, $U = -\mu B$ and $\tau = 0$.
4. **Square loop.** $L_1 = L_2 = 0.45$ m are the loop's side lengths, $I = 0.65$ A its current, $B = 0.65$ T the field in the loop's plane. $\tau = IAB = 0.0856$ N $\cdot$ m.
5. **Hall voltage.** $I = 21$ A is the wire's current, $B = 2.1$ T the field, $n = 8.34 \times 10^{28}$ m$^{-3}$ the free-electron density of copper, $d = 2.588$ mm the wire's diameter (the width the voltage is measured across). $\varepsilon = 4IB/(ne\pi d) = 1.62$ $\mu$V.

**HW 12.1 (fields from wires, forces between wires)**

1. **Four wires on a square.** Each carries the same current and is the same distance from the center. Out-of-page wires on one side, into-page wires on the other, so $B$ at the center points **north**. Draw each wire's field at the center.
2. **Equilateral triangle, opposite currents.** $d = 2.7$ cm is both the separation of the wires and the distance from each wire to P. $I_1 = I_2 = 3.5$ A. Each wire alone gives $B_1 = \mu_0 I / 2\pi d = 2.59 \times 10^{-5}$ T. Two equal fields 120° apart sum to $B_1$, pointing up.
3. **Two perpendicular wires.** Both carry $I = 4.5$ mA. $d_1 = 45$ cm and $d_2 = 5.5$ cm are distances from the points to the wires. All fields are along the page normal, so add them as signed numbers. Point A: $1.44 \times 10^{-8}$ T into. Point B: 0. Point C: $1.84 \times 10^{-8}$ T into.
4. **Parallel wires.** $d = 0.091$ m is the separation, $I_1 = 0.312$ A and $I_2 = 0.683$ A are the currents. $f = \mu_0 I_1 I_2 / 2\pi d = 4.68 \times 10^{-7}$ N/m, attractive. $w = (\mu_0 I_1 I_2 / 2\pi) \ln 2 = 2.95 \times 10^{-8}$ J/m is the minimum work per length to double the separation. The ln comes from $\int_d^{2d} dr/r$, where $r$ is the separation while you pull them apart.

## Common mistakes

- Dropping the sign when reversing the cross-product order.
- Mixing up the two right-hand rules (force versus field around a wire).
- Adding forces or fields as plain numbers when they point different ways. Use components, or signed numbers if all point along the page normal.
- Not converting cm or mm to m.
- Forgetting a negative charge flips the force.
- Using $\sin\theta$ with the wrong angle. In $\tau = \mu B \sin\theta$ it is the angle between $\vec{\mu}$ and $\vec{B}$, not between the plane and $\vec{B}$.